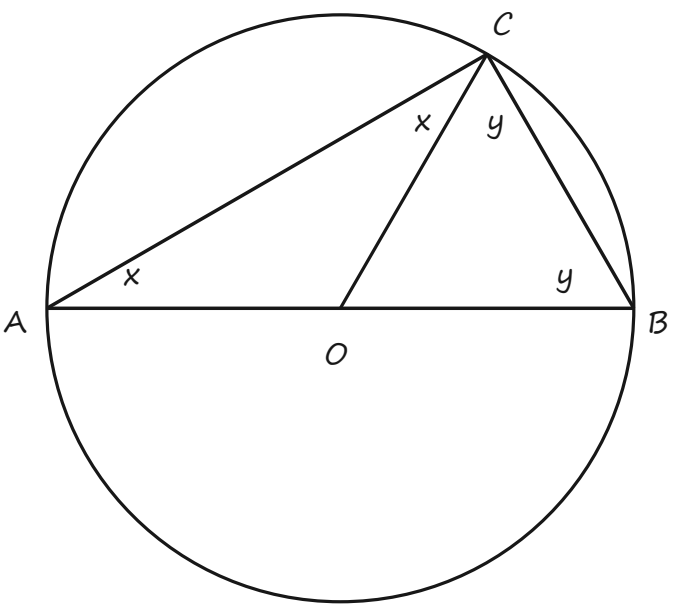
21 A, B and C are points on the circumference of a circle, centre O.

AOB is a diameter of the circle.

Prove that angle ACB is 90°
You must **not** use any circle theorems in your proof.



Join OC. OA = OC = OB because they are radii.

1 mark
Two isosceles triangles share OC.

Let angle OAC = angle ACO = x.

Let angle OBC = angle BCO = y.

In triangle ABC: x + y + (x + y) = 180°.

1 mark
Triangle angles sum to 180°.

Hence 2(x + y) = 180°, so x + y = 90°.

1 mark
Halve both sides.

Angle ACB = x + y = 90°, as required.

1 mark
Base angles of an isosceles triangle are equal; no circle theorem used.

(Total for Question 21 is 4 marks)

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